

Capstone Proposal: CueTracer

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Executive Summary

The use of audio cues within video productions is crucial for enhancing the viewer's experience and conveying intended emotions or messages effectively. However, the process of listing the files used can be tedious and prone to human error. This proposal introduces an innovative solution: CueTracer. My application aims to automate the process of identifying, summarizing, and listing audio cues used in tv spots and trailers, enhancing our post-production workflow and saving countless hours of work.

To implement this application, I intend to containerize the necessary components and use Docker as the platform to deploy them. This will allow me to develop the individual parts in an isolated environment without having to worry about dependency collision or operating system compatibility and to pivot to a different technology for specific modules if the need arises. Further, the experience gained from working in a container environment will serve me well as I continue to develop other applications in the same work environment and in a general professional sense.

I anticipate the amount of time saved from the implementation of CueTracer within our department will lead to further exploration of what workflows could be enhanced via automation. This is one of the reasons I have decided to implement CueTracer via multiple containers as this will allow components to be shared between applications, for example the database component could be used again by other workflows. While this application is being developed to serve the needs of our department and tailored to our infrastructure, I see the possibility that other post-production facilities might have interest in the product. If that is the case, deploying this in another facility should be made much easier due to the container design.

Table of Contents

Capstone Proposal: CueTracer	1
Executive Summary	2
Introduction/Background.....	4
Project Name and Description	4
Problem and/or issue in technology	4
Solution to the problem and/or issue in technology.....	4
Environmental Scan/Literature Review	5
Stakeholders	5
Ethical Considerations	6
Legal Considerations	6
Final Deliverables.....	7
Timeline/Resources	8
Timeline	8
Resources Needed.....	8
References.....	11

Introduction/Background

The goal of my project, CueTracer, is to automate the process of listing audio cues utilized in media productions. This proposal will delve into the specifics of CueTracer, addressing the challenges it seeks to overcome and the solutions it offers. Additionally, I will explore existing projects in this domain, identify our stakeholders, and discuss the ethical and legal considerations surrounding our endeavor.

Project Name and Description

CueTracer will be a server based application that is able to take an AAF(Advanced Authoring Format) file and generate a list of audio files used in the sequence. This will allow our music clearance department to receive automated reports of exactly which cues were used, how many times they were used, and the total duration for which they were used.

Problem and/or issue in technology

When an editor is cutting a tv spot or trailer, they use a variety of audio files to enhance the production quality. Each spot usually includes a mixture of dialogue, music and sound effects. For legal reasons we must account for each piece of licensed music or sound effect used in each spot. Currently this is done manually, with someone reviewing the timeline and making a list of each cue and its duration. To complicate matters, sometimes editors change the name of the audio file in the editing program they are using to suit their needs. This process is tedious, time consuming and fraught with the potential for human error.

Solution to the problem and/or issue in technology

CueTracer will remove virtually all of the manual work, and eliminate the human error element of this process. Using an algorithm called a Fourier Transform, CueTracer will generate audio fingerprints of our entire music and sound effect library, storing these fingerprints in a

database. From the embedded audio AAF file, the application will generate audio fingerprints of each cue used in the given sequence, and will then search the database to find the matching stored fingerprint thus finding the audio file the cue came from. It will build a list of all the matches and, using a third-party library, format this list into an Excel document that is formatted in the way our clearances team has dictated.

Environmental Scan/Literature Review

CueTracer is an incredibly niche product, and to my knowledge there is nothing else out there that can accomplish this goal. There are other projects that are similar, and I am consulting some of them for guidance and to expand my knowledge base around the topic. One in particular is called dejavu(Worldveil, 2020). It allows users to build a database of digital song fingerprints and to find matches on a song by song basis, as do other similar products. Where CueTracer will differ is in its ability to source the audio cues from a given AAF and to produce a final excel report of all matched audio cues.

Stakeholders

The primary stakeholders for CueTracer will be those working in our music clearance department, and the assistant editors that turn over final edits to them for listing. These users currently spend a considerable amount of time manually listing cues in finished spots. I am focused on creating an application that is stable, reliable, and generally free from user intervention, which should eliminate a great deal of manual work for these users. As the developer, I am also a stakeholder and apart from the satisfaction of producing a usable application that makes my co-workers lives easier, I will benefit by demonstrating increased

value to my employer, increasing job security, and hopefully transitioning to doing more development work than my current role allows.

Ethical Considerations

There are a couple things to consider ethically with the deployment of CueTracer. The most glaring issue is job replacement. While the current workflow requires tedious manual labor, those hours are filled by a person. With any automation, the fear is that human jobs will be consumed. From my experience, this is not the case in our department. There is plenty of work to do, and so if we can shift laborious tasks from humans to automation, it will free up those hours for employees to focus on other aspects of their job. The second ethical consideration is regarding accuracy. As a company, we license our sound library and for some of these audio files we pay based on how much they are used. It is important that CueTracer reports an accurate tally of the amount of time each cue is used.

Legal Considerations

CueTracer will be used in an isolated, air gapped, private work environment. The only legal issue I foresee is the use of third-party libraries by a large, for profit company.

Project Goals and Objectives

Goals
Produce a working version of CueTracer that is available for my coworkers to use.
Be able to scan our music library and fingerprint any missing songs in less than five minutes.
Be able to list a full length(2:30) trailer in under five minutes.

Achieve 100% accuracy on cue matches, given the audio file is in the library.
Objectives
Build and deploy the needed docker containers to implement CueTracer.
Write a command line interface for database maintenance.
Implement a watch folder system to monitor for new AAF files
Write documentation for end users, specifically Assistant Editors

Final Deliverables

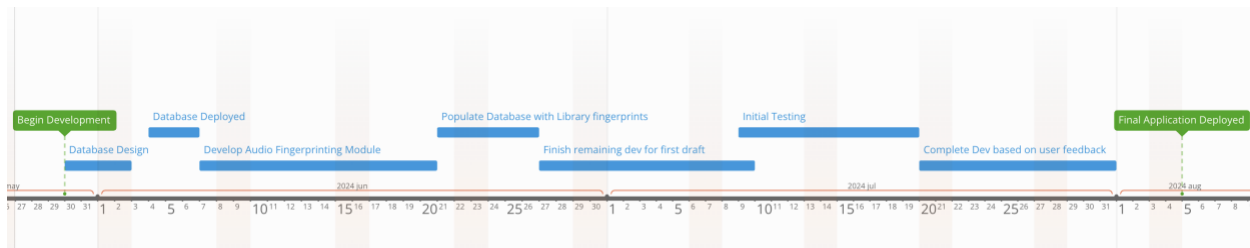
For my final deliverables I will produce a set of docker containers running on a local linux server. This set of docker apps will be able to maintain the database of audio fingerprints as well as process aaf files into excel files formatted in the way our clearance department has asked for.

Approach/Methodology

My approach will follow the Agile development process. While I will be the sole developer, I will engage regularly with the end user stakeholders to assure their requirements are fully understood and achieved. While I am currently not planning on developing a graphical user interface, this is a place where user feedback might lead me to adjust my plans. I will schedule regular meetings that are considerate of their work schedules.

Timeline/Resources

Timeline



Resources Needed

The only resource my project should require is a suitable Linux server to host my applications, which will be supplied by my workplace. I will also have a self hosted server at home so I can do development work when I am not at work. While I don't believe my application will be very compute intensive, that may change when I start running tests on the creation of audio fingerprints. I am prepared to deploy more compute resources if this is the case.

Platform

I will be developing this application using Docker as the platform. Keeping my application container based will allow me to develop it as separate modules that work together. It also allows me to pivot to a different technology if need be, for example if I decide a different type of database would better suit my needs, I can simply spin up the database in a container and point my applications database uses to the new container. For the database maintenance features, I will develop this in python and deploy it as a command line interface. The processing of aaf files into excel files will be handled by a python application that monitors a folder for new aaf files.

Risks and Dependencies

Risks

There are a few risks that could jeopardize the completion of the project. While I fundamentally understand the idea of Fourier Transforms, I have no working knowledge of them. Our music library is extensive, and there is a possibility that the time it takes to fingerprint all these files will be prohibitive. The other risk is that I am the only developer so if I am not able to grasp and implement a piece of the system, there is nobody else to help.

Dependencies

Dependencies for CueTracer start with a working database. Once the database is up and running, using Fourier Transforms to generate audio fingerprints will allow me to build a database of our music library and the audio files we are trying to find matches to. Once we can generate audio fingerprints, we can write an algorithm for finding matches in our database. Finally, once we are able to find matches, we will need to write the module that generates the excel file from a given aaf file. For the final implementation, we will need to develop a directory monitoring service that will begin processing as new files are put in the target directory.

Testing Plan

To test CueTracer I will enroll my coworkers, who will ultimately be the end users of the final product. To test the database management component, I plan to have our systems engineer test the module, as well as help monitor the conversion of our music library and population of the database. For the remaining testing, I will have our assistant editors generate aaf files, and our clearances department verify the excel files are formatted properly and captures all of the used audio files.

Members/Division of Responsibility

Since I am the only developer, there is no division of responsibility. I will complete all work on CueTracer.

References

Worldveil. (2020). Dejavu [Software]. GitHub. <https://github.com/worldveil/dejavu>

Appendix